

Can AI Generated Ambrotype Chain the Aura of Alternative Process?

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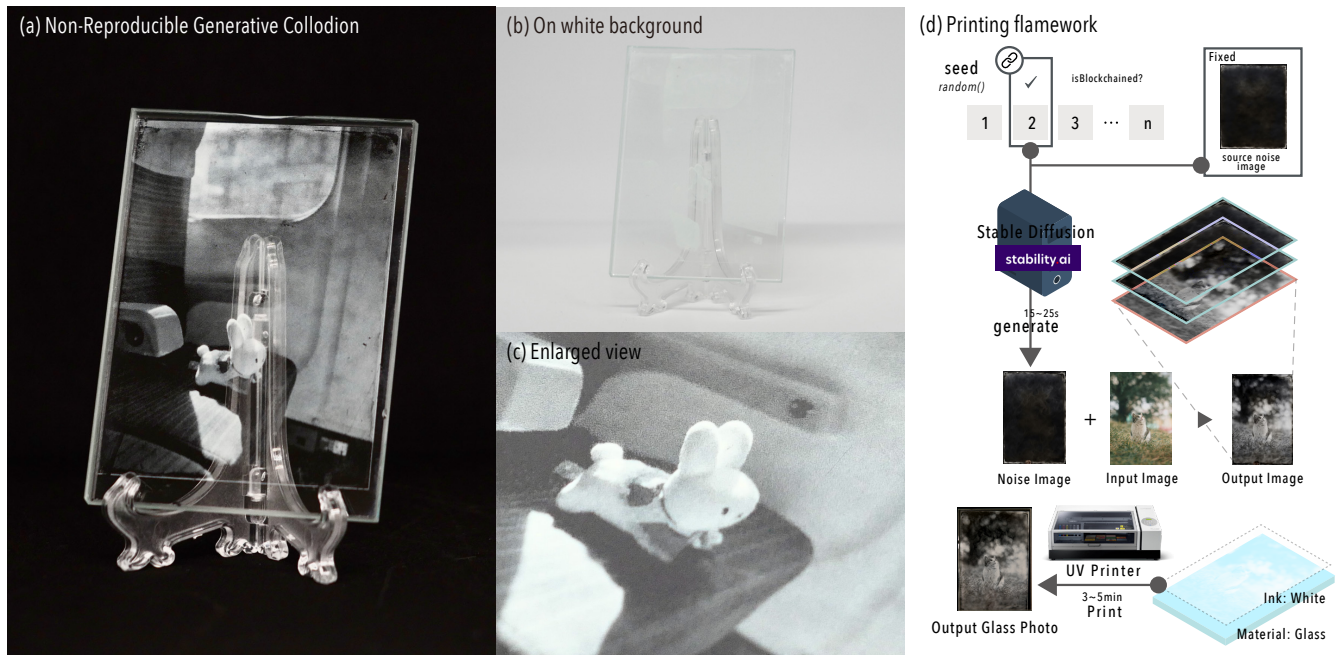


Figure 1: The Non-Reproducible Generative Collodion process: This figure presents both the delightful final images and the system that produces them. The process integrates traditional wet plate collodion photography, especially ambrotype, techniques with modern digital fabrication technologies. Key steps include image transformation via generative AI to emulate wet plate textures, ensuring uniqueness through NFTs, and final printing on glass plates using UV inkjet printing. The results showcase the charming blend of handcrafted aesthetics with the replicability and scalability of digital technologies, exploring new meanings of aura in contemporary photography.

Abstract

"Non-Reproducible Generative Collodion" integrates traditional ambrotype with digital fabrication, using generative AI and NFTs to ensure each photograph's uniqueness. This method redefines "aura" in the digital art era, offering new perspectives on photographic art by combining classical techniques with modern digital advancements, as supported by expert feedback.

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CCS Concepts

• Applied computing → Media arts.

Keywords

Aura, Replication, Alternative Photographic Process, Wet Plate Collodion, Ambrotype, Generative AI, NFT, Digital Fabrication

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1 Introduction

What does "reproduction" mean in the modern age, where computers are highly advanced? Looking back through art history, works of art have traditionally been labeled with information about their

creator, date, and place of creation, ensuring their originality and uniqueness [Benjamin and Jennings 2010]. Such works, born from specific locations, individuals, and temporal contexts, have been known to possess an "aura", evoking feelings of awe and sublimity among people.

Approximately 200 years have passed since the invention of heliography, which is often considered the birth of photography. As photography, a representative of technological reproducibility gained recognition as an art form, the tradition of including edition numbers on copperplate prints continued, becoming an indispensable practice in contemporary art. Limiting the edition has preserved the uniqueness and unique value of each work. However, compared to the aura of paintings, this aura is said to have diminished.

In recent years, many art pieces exist in non-material forms. For digital art, technologies like NFTs (non-fungible tokens) establish ownership and uniqueness, replacing edition numbers and certificates of authenticity typically found on physical works [Nadini et al. 2021]. Generative AI, using language, programming, and seed values, can produce identical outputs. These generated items are not inherently unique but exist as reproducible entities. In this context, how does the aura function?

The discussion surrounding "aura and reproduction" holds significant technical and cultural importance within art and technology.

This paper focuses on the debate about the aura in wet plate photography, especially ambrotype, created using chemical processes, and in photographs reproduced using digital fabrication technologies. Walter Benjamin, photographic-native, argued that photography eradicated aura. Before Benjamin, however, photographs could not be printed identically each time (Figure 4). He acknowledged this in his work "A short history of photography." [Benjamin 1972]. In contrast, contemporary photography allows for consistent reproduction. The same applies to generative AI. However, how can aura exist in contemporary reproducible photography? Reconsidering photography through techniques that introduce randomness and uncertainty, similar to wet plate collodion photography, alongside reproducibility offered by generative AI, is posited to bring forth new cultural critiques regarding the fundamental question of "what constitutes an original" within the digital art community.

This paper makes the following contributions:

- Examination of photography techniques and the concept of aura within the graphics community in 2024, referred to as the era of ultra-reproduction technology.
- Proposal of a method to create photographs resembling ambrotype using digital fabrication and blockchain technologies, termed "Non-Reproducible Generative Collodion".
- Insights, considerations, and critiques regarding the aura users perceive in traditional ambrotype and Non-Reproducible Generative Collodion.

This paper critiques the relationship between photography and aura within digital art, connecting past and present to explore the essence of photography and usher in a new era.

2 History and Value of Wet Plate Collodion Photography

2.1 Historical background of wet plate collodion photography

Wet Plate Photography, widely used in the mid-19th century and commonly referred to as itinerant photography, was famous for portrait photography [Osterman 2008]. Among wet plate photographs, ambrotypes, which use glass plates as a substrate, are still pursued passionately by enthusiasts because of their beautiful finish. Wet plate photography combined two advantages: the high sharpness of the daguerreotype [Barger and White 2000] and the ability to produce multiple copies like the calotype [Miller 2003]. The advent of this technique made it possible to provide clearer images in a reproducible form, greatly contributing to the development of photographic technology. At that time, in the field of commercial photography, new tools, processes, and techniques for taking better photographs more quickly were highly valued. Just as people today do not know in detail how smartphones are made, people of that era did not understand how photographs were created, so photography was truly a magical existence [Persinger 2014]. As a result, many people eagerly sat nervously in front of the camera, rushing to experience this magical device to obtain photographs of themselves and their families for the first time. In this way, the advent of wet plate photography became a turning point for photography to be widely treated as a medium. In terms of technical significance, it achieved high resolution at the level of silver atoms, making it possible to capture details that previous techniques could not express. In terms of cultural significance, photography came to have value as a permanent recording medium, and it became established in society as a means to preserve people's memories and history.

2.2 Characteristics of the ambrotype and its artistic value

As shown in Figure 2, this production process is complex and delicate. It requires a high level of expertise influenced by many uncontrollable factors.

A widespread mistake in wet plate collodion photography is that fogging or partial peeling of the image often occurs (Figure 3). Addressing these issues requires years of experience and intuition to improve quality [Ellis 1856]. Thus, wet plate collodion photographs emphasize their uniqueness through differences attributable to region, creator, or era. This uniqueness and the rigorous creation process form the aura of these works.

3 The Aura in the Era of Ultra-reproduction Technology

Modern viewers are used to pixels. As photos and videos have digitized, they are now made up of tiny pixel units, which has changed how they are appreciated and evaluated. This digitization has seemingly erased the aura and diminished the value of photos as material objects. With the vast number of digital photos taken and shared daily [Miller and Edwards 2007], each photo's unique

The Production Process of Wet Plate Collodion Prints

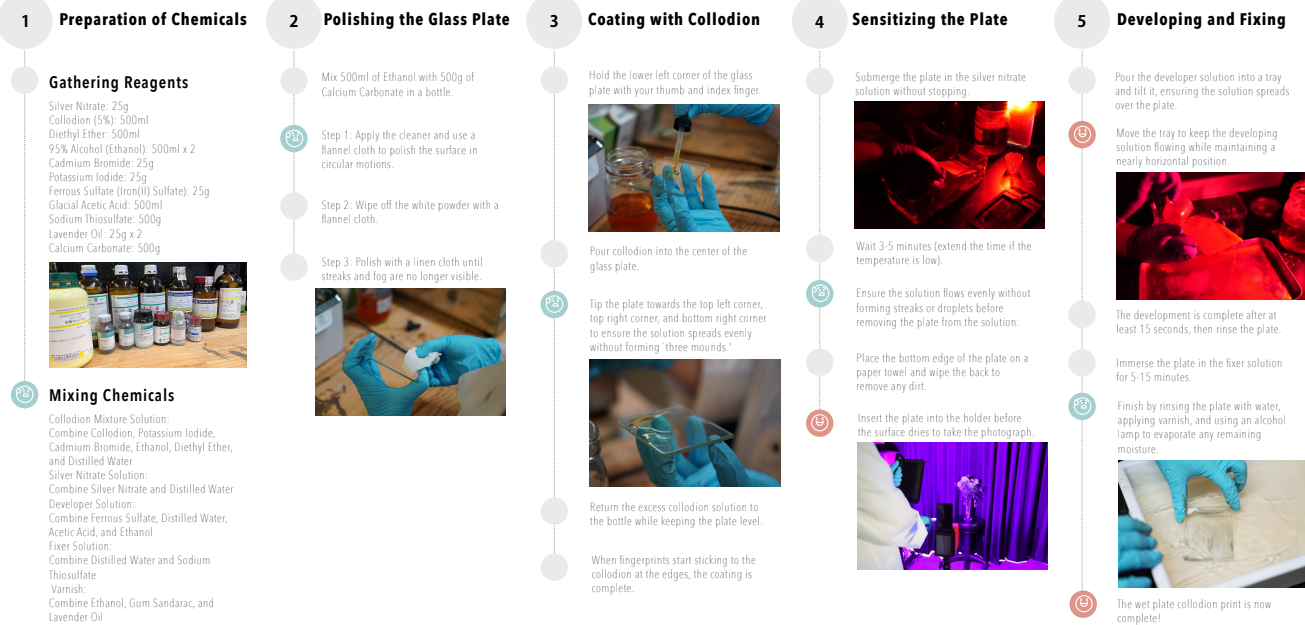


Figure 2: The wet plate process, famous for its beautiful finish, involves complex and delicate tasks such as precise chemical handling, adaptation to environmental conditions, and experience-based judgment. It requires high craftsmanship, with methods varying by region and era.



Figure 3: Fogging or partial peeling of the image is a common mistake in wet plate collodion photography. These errors result from the quality of chemicals, differences in absorbed ultraviolet light, or insufficient sensitization or fixing.

value has faded, making them mere parts of a larger information landscape¹.

¹<https://www.bbc.com/news/technology-54919165>

Considering this, what can be said about the art of the "ultra-reproduction technology era" in modern times? Just as printing a single photograph involves many steps, current research in digital fabrication technology involves efforts to print various objects from data. Although these efforts are primarily engineering pursuits, one might argue that the joy of creation itself attracts engineers to these endeavors.

For instance, in the realm of everyday dining, food 3D printing applications can output dishes with predetermined calorie counts while maintaining their appearance [Lin et al. 2020] or produce visually intended jelly desserts with precision [Miyatake et al. 2021]. These technologies enable the output of actual food and its aesthetic appearance from data. Similarly, in personal projects like crafting, computers can assist in producing desired patterns, such as sewing stuffed toys by merely cutting and stitching fabric according to shapes determined by computers [Mori and Igarashi 2007] or using 3D printed looms to create complex weavings with computer guidance [Hudson 2014]. There are also larger-scale applications, such as using portable plastic extruders to fabricate wireframe prototypes of furniture directly [Agrawal et al. 2015] or designing furniture computationally and assembling it from laser-cut pieces [Saul et al. 2010].

In this way, computers facilitate and automate complex, iterative tasks while respecting manual craftsmanship's human appeal. Contrary to the simple and efficient production processes familiar

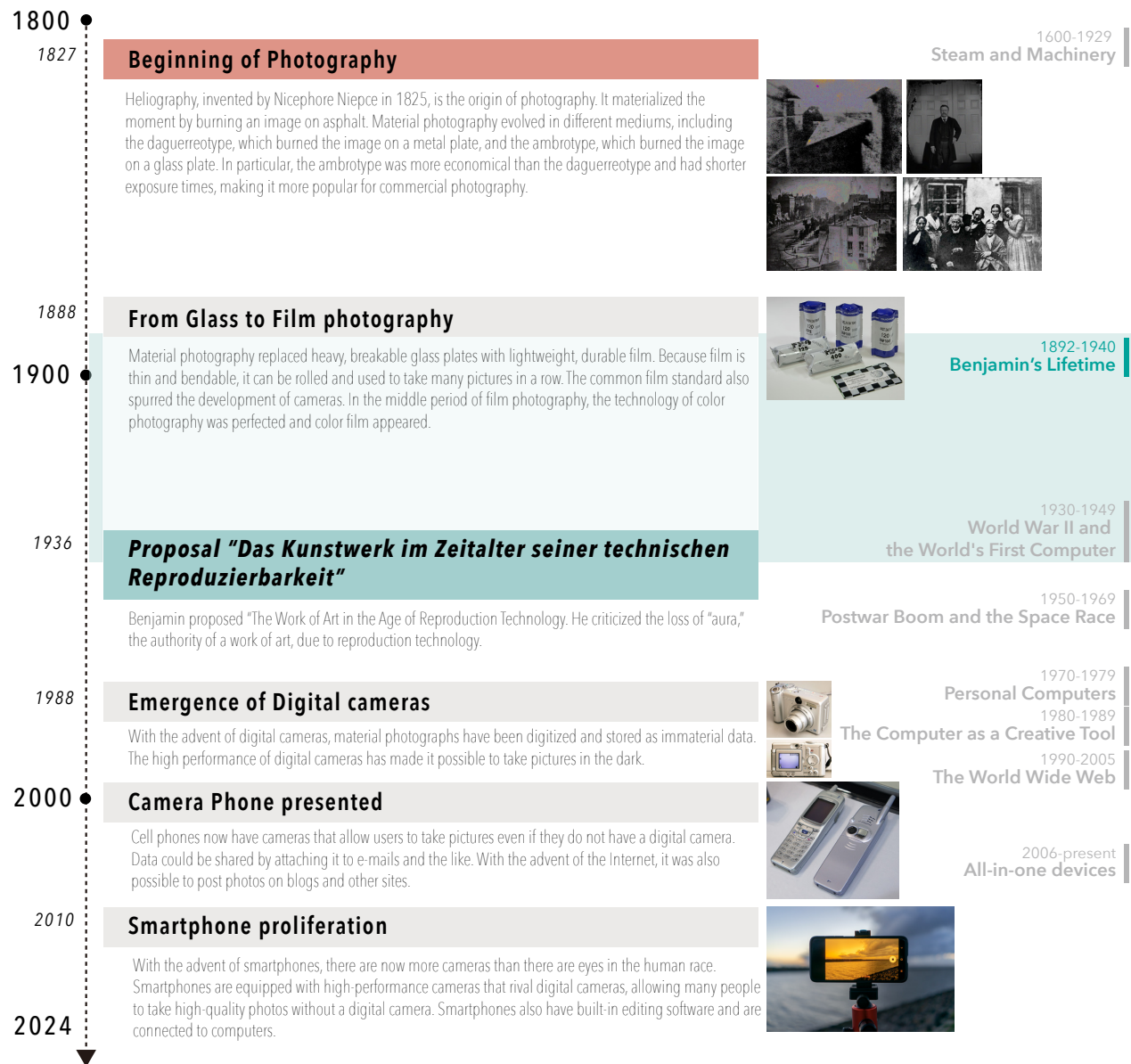


Figure 4: Timeline of Reproduction Technologies from the Proposal of Aura to the Present. Before Walter Benjamin argued that photography eradicated aura, photographs could not be printed identically each time. Contemporary photography and generative AI allow for consistent reproduction, raising questions about aura and originality. Techniques introducing randomness and uncertainty in wet plate collodion photography offer new cultural critiques.

to contemporary people, the value found in these manual methods can be considered as the source of Aura for modern individuals.

4 Proposal of Non-Reproducible Generative Collodion

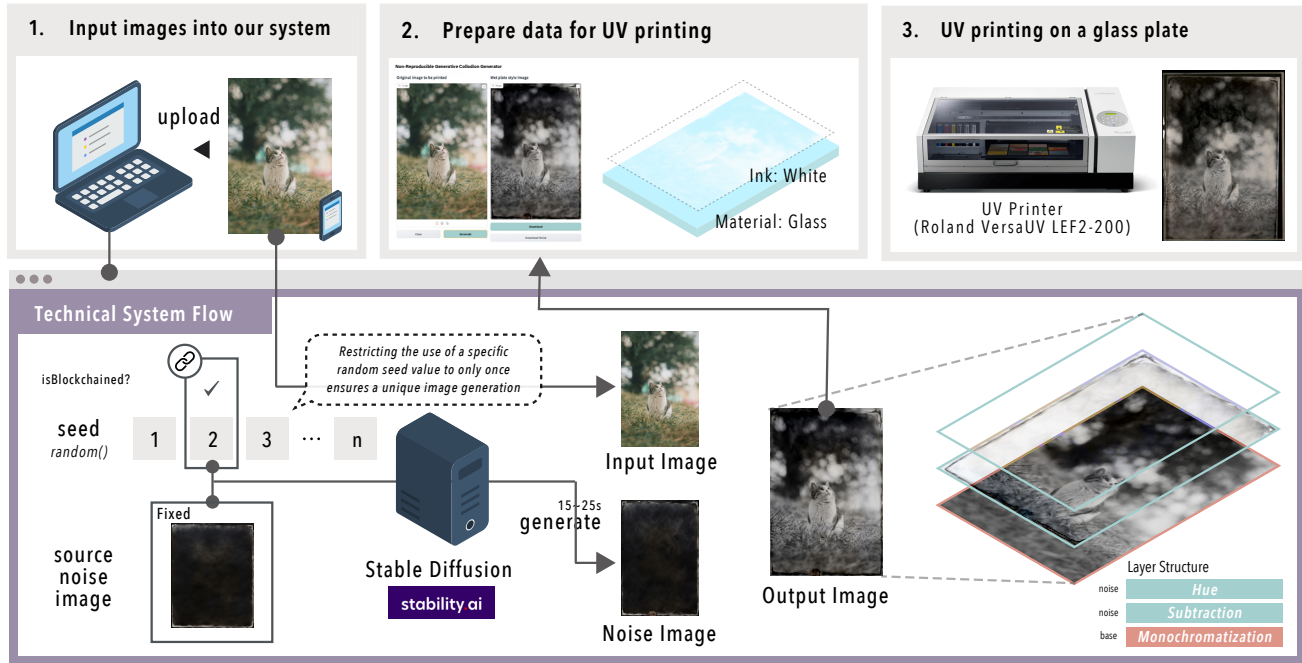
We proposed "Non-Reproducible Generative Collodion," a method to create glass photos using digital fabrication technology.

4.1 Design Principles

To develop a new photographic technique that chains the aura by combining classical photographic techniques with modern digital fabrication, we established three design principles for it:

- maintaining the experiential nature of ambrotype
- reducing the steps required compared to traditional methods
- preserving the uniqueness of each photograph

(a) Overview of our method



(b) Image with different textures: 3 examples



Figure 5: The system for creating Non-Reproducible Generative Collodion. (a) The system applies unique image transformation to the input image, adding one-of-a-kind wet plate-style noise using Stable Diffusion. Images can be downloaded and transferred to a UV inkjet printer and printed on glass plates for production. (b) The different images generated by random seed values: random seed values are recorded as NFTs on the blockchain to prevent re-use, and the system verifies their uniqueness at the time of image generation.

For the first principle, we focused on two key aspects: the quality, whereby images appear as negatives on a white background, positives on a black background, and glass as the medium. These were achieved by printing the positive image data onto a glass plate with white ink.

For the second principle, we utilized digital fabrication technology, specifically a UV inkjet printer capable of printing on glass.

For the third principle, user input images are transformed into a wet plate style using a generative AI engine. Recording the seed

values and using NFTs ensures the uniqueness of the resulting images.

4.2 Development Process

Based on the design principles outlined in 3.1, a system was developed to create Non-Reproducible Generative Collodion (Figure 5(a)). This system applies a unique image transformation to the input image. This transformation adds one-of-a-kind wet plate-style noise to the input image using Stable Diffusion. The uniqueness of the

transformation is guaranteed by Stable Diffusion and NFTs. Stable Diffusion is a generative AI that produces the desired image by denoising Gaussian noise in the latent space based on a random seed value. In Stable Diffusion, the random seed value determines the generated image. Restricting the use of a specific random seed value to only once ensures a unique image generation, thus translating the concept of uniqueness from the physical world into the digital world (Figure 5(b)). The system records the used random seed value as an NFT on the blockchain, preventing its reuse. During image generation, the system verifies that the random seed value has not been previously recorded on the blockchain, ensuring the uniqueness of the generated image. See our supplementary materials for instructions on printing on glass.

4.3 Evaluation of Final Outputs

In our framework, we create glass works using generative AI and NFT as core technologies. The results demonstrate variations in texture depending on the production order and differences in hue when viewed against a black background (Figure 6). Traditional ambrotype and our Non-Reproducible Generative Collodion show natural blending textures, though improvements are needed in hue consistency.

5 The Position of Aura in Photography and Non-Reproducible Generative Collodion

Regarding photographic techniques known as alternative processes, they were classified using a four-quadrant matrix with the axes 'organic-mechanical' and 'handcrafted-industrial'. When alternative processes are plotted on this matrix, they appear as shown in Figure 7. In existing processes, there are very few instances of photographs that are both organic and industrial. The proposed Non-Reproducible Generative Collodion achieve this through computer technology. The synergy between the organic aspects of generative AI and NFTs and the industrial attributes afforded by digital fabrication technology demonstrates a new position for photography that has not been possible.

6 Expert Critiques and Feedback

6.1 Methodology of Expert Interviews

This study conducted semi-structured interviews with ten photography experts (9 males, 1 female) to evaluate the production method and aesthetic appearance of Non-Reproducible Generative Collodion, as shown in Table 1. Participants experienced both traditional and Non-Reproducible Generative Collodion and shared their impressions. Interviews were done either face-to-face or online based on participant preference. Five (4 males, 1 female) chose face-to-face, where they could handle and observe the subjects freely in a quiet environment. The other five (all males) participated via Zoom, examining high-quality images in a quiet space. In both formats, participants were briefed on the experiment's purpose and procedure and then compared both types of glass plate photographs, discussing their impressions.

6.2 Analysis of Feedback on Face-to-Face Implementation

First, there was a sentiment that Non-Reproducible Generative Collodion felt contemporary. Comments included, "Feels like a modern expression [P3]", "Color and clarity are modern [P3]", "The image of the wet plate has been overturned [P3]", and "Feels like a crossing between digital and analog [P10]". Positive evaluation included, "It is a beautiful ambrotype, unlike anything I have seen before [P5]", "The mid-tones and highlights are beautiful [P5]", "The highlights are excellent [P5]", and "There are irregularities, but they feel inherent to the wet plate [P3]". However, there were several critiques, primarily concerning color perception. Opinions included, "The color is different from wet plate; it is not brownish [P3]". Other comments related to white-black conversion or the color of silver: "This is the color of monochrome print [P7]", "There is no metallic sheen [P5]", "The white parts in the highlights feel different from silver [P8]", "The texture looks like digital dots [P7]", "The resolution feels different [P10]", and "It feels similar to a Polaroid [P7]". Experts also made predictions about the production method. Analog perspectives included, "Uniform gelatin with color removed by chemicals [P5]" and "Duplicated from digital data [P7]". Digital method predictions included, "Using a printer [P6]", "Silkscreen [P8, P10]", and "UV printer [P9]". Regarding the production method's new direction, praises included, "Very highly evaluated as a new direction [P7]" and "Feels like a process with presence [P3]". Positive traits of the generic method were, "It is appealing that anyone can do it [P3]" and "Valuable for creating large pieces unachievable in traditional wet plate [P3]". Feedback also acknowledged the system's uniqueness: "It is good, ensuring uniqueness so that the same item cannot be made again [P5]" and "The unpredictability of how the fuzziness appears is amusing [P8]". Some comments identified areas for improvement, such as "For those who consider the noise as a defect, it is a bit of an issue [P5]". Others appreciated the noise: "The noise is a charming aspect of wet plate [P5]", likening it to works by Sarah Moon or Witkin [P7]. Some saw overlapping generative AI with light-captured photos as conceptual and artistic: "When AI-generated images merge with actual light-captured photos, it becomes more like art [P7]". Comparisons with traditional wet plate photographs highlighted specific traits associated with their aura: "The sense of manual work [P3, P10]", "The human essence [P3]", "The material feel of the glass [P3, P8]", "The material feel of the film [P10]", "The metallic sheen [P5]", "The color reproduction where reds appear black and blues appear white [P7, P8]", and "The sense of time and space present during shooting [P8, P9]". Unique advantages of Non-Reproducible Generative Collodion included: "The ability to change the color without altering the chemicals is good [P3]"; "The highlights are more beautiful than traditional wet plate photographs [P5]"; and "The generic versions appear prettier [P7]". Summarizing the interview reflections: "The more generic it is, the further it diverges from the traditional wet plate, which feels interesting [P3]" and "In an era overflowing with AI-generated images, treating photos as physical objects is delightful [P5]" highlighted the charm of traditional wet plate and re-evaluated the flexibility afforded by computer technologies.

(a) Comparison of traditional and our methods



(b) Difference in printed generated noise



(c) Design session of Non-Reproducible Generative Collodion



Figure 6: Comparison of traditional ambrotype and Non-Reproducible Generative Collodion. Using the same image results in texture variations depending on the production order. Both methods exhibit textures that blend naturally, providing a harmonious appearance, but differences in hue are noticeable, especially against a black background.

Positioning Map for Pre-digital Photography

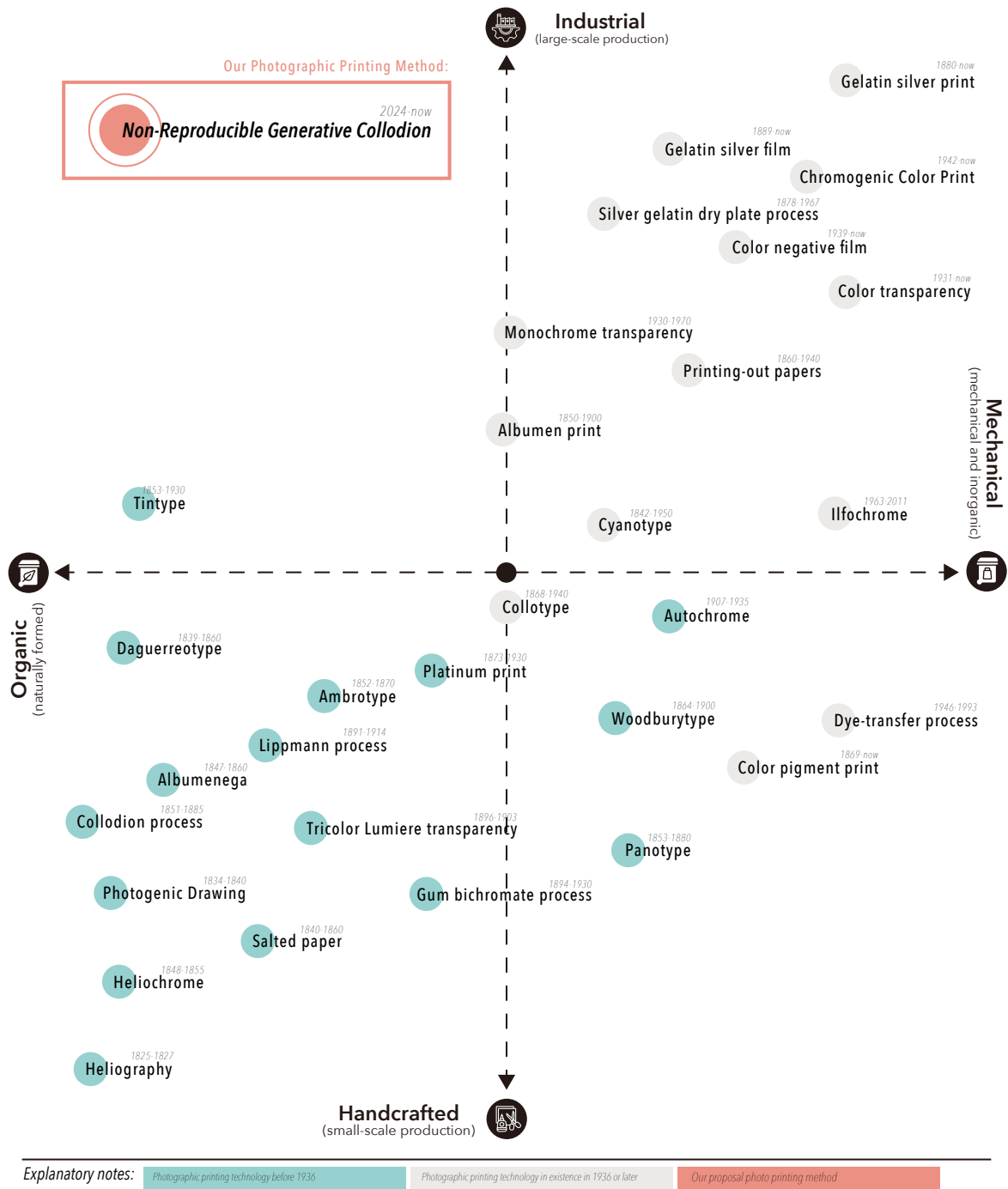


Figure 7: Classification of photographic techniques using a four-quadrant matrix with 'organic-mechanical' and 'handcrafted-industrial' axes. The matrix shows the positioning of various alternative processes, highlighting that there are few instances of photographs that are both organic and industrial. Non-Reproducible Generative Collodion occupies a new position by combining generative AI and NFTs (organic) with digital fabrication technology (industrial), representing a unique integration in the field of photography.

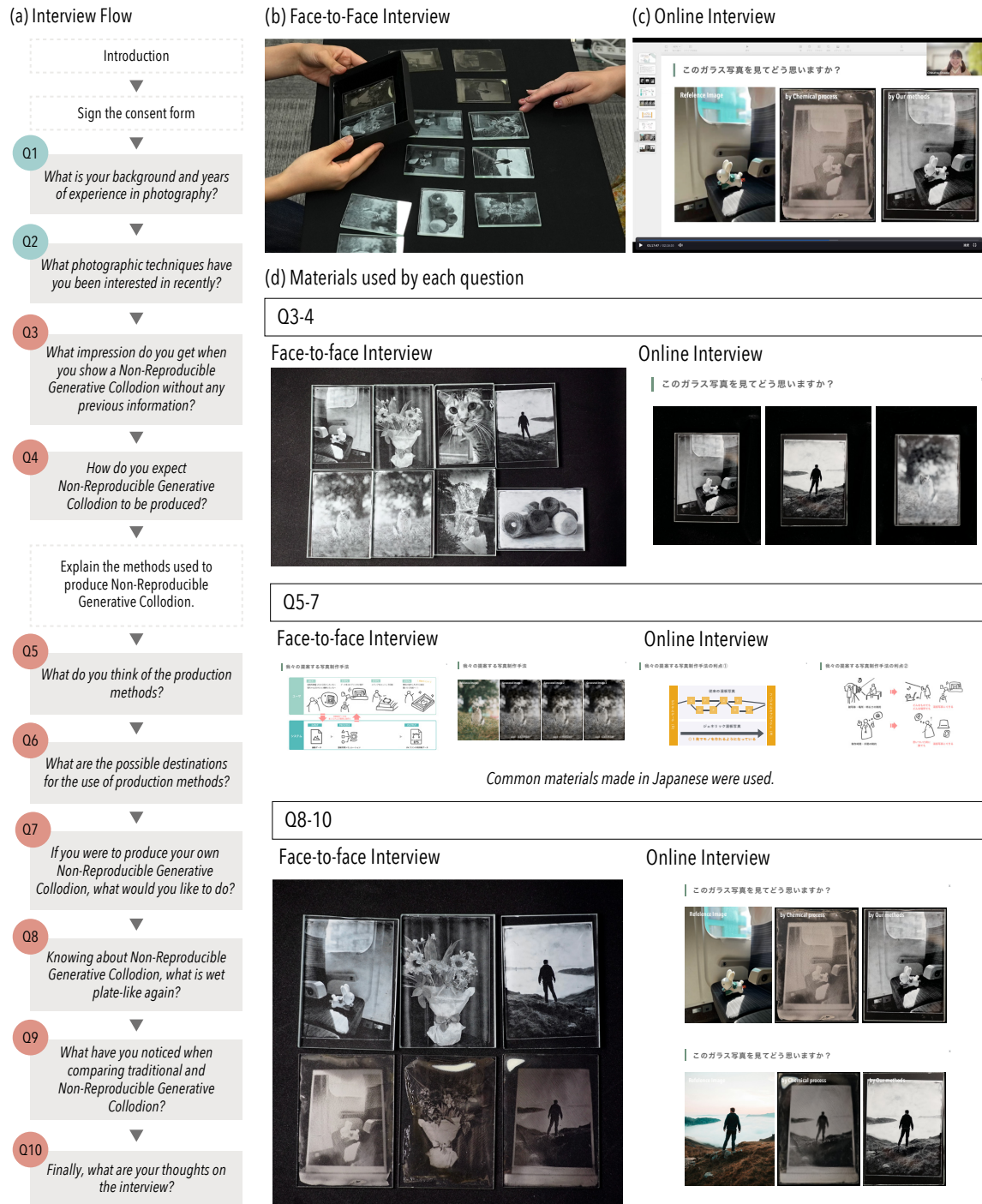


Figure 8: Interview questions and itemised materials used: interview items were used both face-to-face and online. The materials were conducted in an environment where the participants could freely pick them up or use a loupe to observe them in face-to-face situations. For online, images taken in orthographic projection were pasted onto slides and shared on screen. The playback time of the pictures was manipulated according to participants' cues.

Table 1: Biographies of photographic experts who participated in the interviews.

Number	Photographic Background
P1	Started film photography at 18, became a photographer after university. Returned to analog, commercialized wet plate. Does family photos, workshops, camera development.
P2	Started at a BW photo lab at 27, went independent 26 years ago. Runs a color and BW lab. Began alternative techniques 15 years ago, wet plate for 12 years.
P3	Practicing monochrome for over 40 years. Includes silver halide, platinum, and alternative techniques. Printer for Leica Ginza, Toyo Award winner, former visiting professor.
P4	Got a camera in university, resumed film photography later, passionate about BW film. Interested in darkroom-less classical techniques, actively involved.
P5	Started with mountain photography, medium format, self-developing. Bored with digital, practices Kallitype, creates daguerreotypes, explores semi-darkroom techniques.
P6	Started a wet plate site as a student. Inspired by camera-collector relative, patented wet plate photo. Practicing for eight years.
P7	Running wet plate business for 11 years. Moved from digital, focuses on wet plate, does albumen occasionally.
P8	Started in university, 40 years in ads and magazines. Learned fine art, large format, and classical techniques. Provides technical support to photo studios.
P9	Started film photography in elementary school, majored in it through graduate school. Now advertising photographer, lecturer, and researcher.
P10	Studied photography at university, 40 years researching platinum printing. Works in preservation framing, gallery exhibitions, and sales.

6.3 Analysis of Online Feedback

First, upon showing Non-Reproducible Generative Collodion without any information, participants collectively remarked, "It feels like wet plate [P1, P2]" but also noted, "I quickly realized it wasn't wet plate [P1]". Reasons included: "It feels somewhat like digital processing [P1]", "Cannot be taken in a normal setting [P2]", "Subjects look modern [P6]", and "Has modern colors [P4]". Experts speculated about the production method. Analog guesses included: "Developed using collodion [P2]" and "Replication using a monitor [P1]". Digital guesses were: "Using lithographs [P4]", "Transferred using UV printer [P1]", and "Transferred using digital negative [P6]". Positive feedback about the production method included: "Achieving what I envisioned in an era when digital fabrication was not widespread; it feels like a dream realized [P1]", "Considering a system that prevents uniformity in app-based results is impressive [P1]", and "This feels like something from the future [P4]". Critiques pointed out areas for improvement: "Color reproduction felt different [P1, P2]", "The experiential aspect is completely missing [P1]", and "(Having limited freedom) constrains the photographer [P4]". Comparisons with traditional wet plate photographs highlighted the significant contribution of the experience to their distinctive aura: "The Non-Reproducible Generative Collodion further emphasizes the importance of the experiential aspect in traditional wet plates [P1]", "The color perception due to photosensitive characteristics is unique to wet plates [P2]", and "The six-second exposure time and the immobility contribute to the aura [P6]". Advantages of Non-Reproducible Generative Collodion included their clarity and detail: "The Non-Reproducible Generative Collodion is clear and easy to see [P1]", and "The ability to capture fair and clean skin tones is good [P6]". Its potential for modern applications was noted: "It is fascinating that photos taken with a smartphone can have a wet plate appearance [P6]", and "Capturing the atmosphere of the

current era makes it interesting [P6]". The vividness and brightness of Non-Reproducible Generative Collodion were praised: "While traditional wet plates are often dark and murky, the Non-Reproducible Generative Collodion are bright and clean [P6]".

6.4 Non-Reproducible Generative Collodion and Aura

Interviews revealed our proposed photographic technique as a viable future form of physical photography (Figure 9). Some experts noted an "indescribable mystery" and remarked the results were "so beautiful that it is hard to believe," suggesting a sensed aura. Despite noted areas for improvement, acknowledging an aura in this "organic and industrial" genre implies that aura in photography cannot be entirely separated from "reproducibility."

These responses stem from the "materiality" of glass and the "clarity of aesthetic appearance" absent in traditional wet plate photographs, highlighting contemporary sensibilities. Additionally, experts familiar with analog methods appreciated the modern approach of creating unique images using an NFT-based system, showing that our ultra-replication technology offers a fresh perspective in photographic art.

7 Discussion

7.1 Three Points That Make Up the Wet Plate Photographic Style and Their Reproduction

Interviews clarified that the "style" of wet plate photography relies on three main aspects: "experiential nature," "color perception," and "uniqueness." We mainly focus on "color perception," which almost all experts highlighted.

Wet plate photography has unique photosensitive characteristics not replicable by simple color-to-black-and-white conversion. For

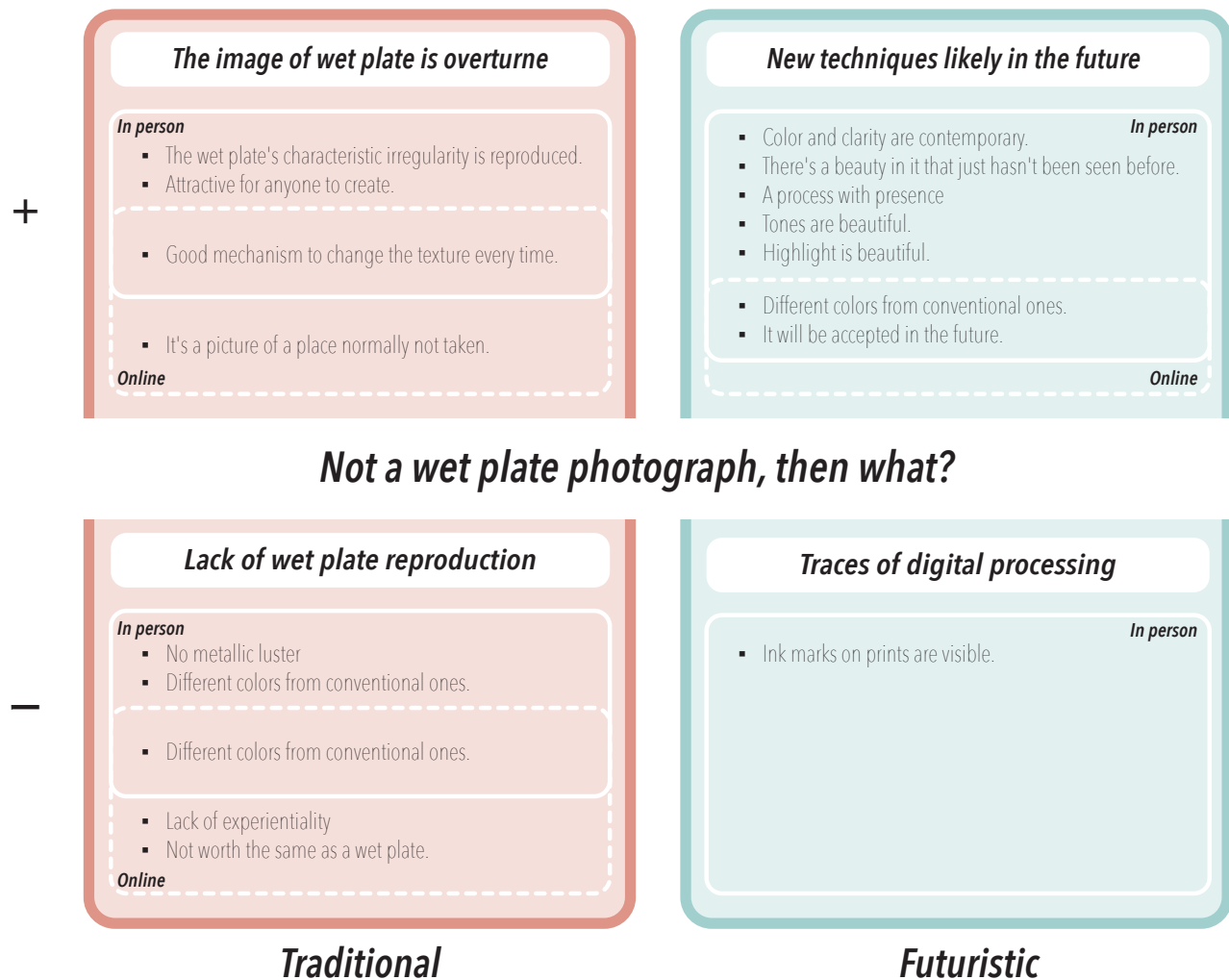


Figure 9: Results of the interviews are summarized and analyzed. The left-hand column describes the content compared to traditional wet plate photography, and the right-hand column describes the content as a new futuristic medium. The upper row shows positive content and the lower row shows negative content. The in-person implementation is indicated by a solid line and the online implementation by a dotted line, with the intersection representing the content noted for both implementation methods.

example, reds and yellows appear darker, while blues and purples appear lighter (Figure 10). However, our system uses basic black-and-white conversion. Experts noticed this difference, perceiving it as a deviation from traditional methods. Non-experts did not mention this, indicating it is mainly a specialist concern.

Wet plate photography has unique photosensitive characteristics not replicable by simple color-to-black-and-white conversion. For example, reds and yellows appear darker, while blues and purples appear lighter. Our system uses basic black-and-white conversion. Experts noticed this difference, perceiving it as a deviation from traditional methods, whereas non-experts did not cite this issue, indicating it is mainly a specialist concern.

UV inkjet printers resulted in a textured appearance due to sprayed ink. Both online and face-to-face participants identified this texture as a factor distinguishing the photographs from traditional wet plate ones. While traditional wet plate photographs have depth, Non-Reproducible Generative Collodion lacked this. Experts noted that rough texture, lack of smoothness, and insufficient image layer thickness contributed. Adjusting UV printer ink thickness and density could better mimic authentic wet plate texture.

Traditional ambrotypes exhibit a metallic sheen due to light hitting silver particles. Non-Reproducible Generative Collodion, printed with white ink, do not showcase this sheen. To address this, experimenting with inks containing silver nanoparticles might be necessary.

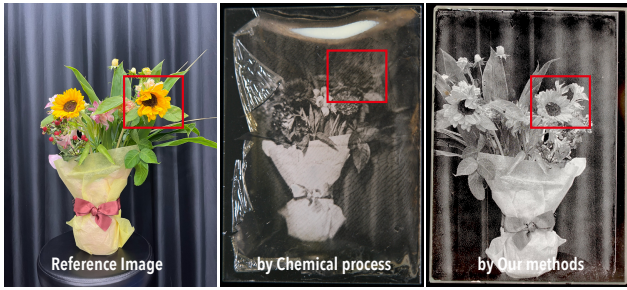


Figure 10: Wet plate photography shows unique photosensitive characteristics where reds and yellows appear darker, and blues and purples appear lighter. These differences are not captured by basic black-and-white conversion.

7.2 Differences in How a Non-Reproducible Generative Collodion Looks Depending on the Photographic Technique Experienced

The interviews in Section 5 revealed that experts' primary photographic techniques influenced their impressions of Non-Reproducible Generative Collodion. Experts in ambrotype emphasized photosensitive characteristics. Those experienced in wet plate techniques compared these with our Non-Reproducible Generative Collodion, which uses simple black-and-white conversion and lacks photosensitive properties. This led some experts to note differences in color gradation and tone. However, some praised the clarity and beauty of the midtones and highlights in our Non-Reproducible Generative Collodion. Additionally, the unique visual property of ambrotypes, appearing as negatives on a white background and positives on a black background, was well received.

Experts working with black-and-white film or paper-based techniques highly valued the tonal beauty and compositional completeness. They praised the clarity and beauty of the midtones in the Non-Reproducible Generative Collodion. Those unfamiliar with wet plate photography did not comment on color perception differences related to photosensitive characteristics.

Experts focused on metal-based techniques particularly valued the metallic sheen of silver, which changes slightly with different viewing angles due to reflected light. One expert noted that this metallic sheen is an unconscious allure of ambrotype. Although not frequently mentioned, this sheen might also unconsciously attract other experts.

7.3 Potential Demographics Where Non-Reproducible Generative Collodion is Acceptable

Potential users captivated by our method fall into two categories. One includes alternative photographers who embrace new techniques fused with digital technologies, showing enthusiasm for innovations in Non-Reproducible Generative Collodion using digital negatives and 3D printers.

The other category includes young, digital-native individuals who are new to glass photography but find film cameras novel. They,

along with younger generations and women, are likely to appreciate ambrotype for its immediacy and simplicity, similar to instant photography. The interviews suggest that Non-Reproducible Generative Collodion would appeal to this demographic, highlighting its potential as a new technique in the future digital art community.

8 Conclusion

In this study, we proposed the "Non-Reproducible Generative Collodion," which integrates traditional ambrotype techniques with digital fabrication technology, and evaluated its production methods alongside the accompanying concept of "aura."

Firstly, we reexamined the relationship between photographic techniques and the concept of "aura" in the contemporary era of ultra-reproducibility. While advancements in digital technology have dramatically improved the reproducibility of photographs, it can be said that the "uniqueness" and "aura" of artworks have diminished. However, in wet plate photography—particularly ambrotypes—the inherent uncertainties of chemical processes and manual craftsmanship contribute to the uniqueness of each piece, preserving their "aura."

In this research, we developed a method to ensure the uniqueness and individuality of artworks within a digital environment by utilizing generative AI and NFTs (Non-Fungible Tokens). Specifically, we employed generative AI to add unique wet plate-style noise to input images and recorded the seed values used during this process as NFTs on the blockchain. This prevents the reproduction of identical works. Additionally, by printing onto glass plates using UV inkjet printers, we provided the tactile quality and viewing experience of physical artworks.

Interviews with experts confirmed that the works generated through our proposed method possess new value distinct from conventional ambrotypes. Notably, the ability to evoke a sense of "aura" and individuality, even while leveraging digital technology, was highly appreciated. On the other hand, areas for improvement were identified, such as color reproduction, texture, and the replication of the silver sheen characteristic of traditional wet plate photography.

From these findings, it is suggested that the fusion of traditional techniques with modern digital technology can unlock new possibilities in photographic expression. Particularly for the digitally native generation and artists seeking new methods of expression, the unique production process and resulting works offered by our method are considered to hold great appeal.

Looking ahead, further technical refinements—such as enhancing color fidelity and replicating the silver luster—are expected to bring our method closer to traditional wet plate photography. Moreover, exploring combinations and applications with other traditional photographic techniques and artistic expressions could lead to new points of convergence between digital art and traditional art forms.

This study is an endeavor to explore new artistic possibilities through the dialogue between classical techniques and contemporary digital technology, illustrating how technological advancements can transform artistic values. We believe it provides fresh perspectives on how human creativity and sensibility may evolve.

In conclusion, the Non-Reproducible Generative Collodion serves as a bridge connecting tradition and innovation, presenting a new method of expression through the fusion of photographic art and

digital technology. We anticipate that, with further research and refinement, this approach will shed new light on the concept of "aura" and the value of artistic works in the modern era.

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